

highly correlated with the stock market, the traditional financial planner advice is inappropriate!

You are now going to object because you know a sixty-year-old stockbroker who lost his job in the Great Recession in 2008 and he held all his wealth in equities. He lost both his job and his wealth at one fell swoop and seemed to (unintentionally) follow the life-cycle investing advice. The life-cycle theory is a relative statement on how allocations change with time: for an individual whose labor income is stock-like, he should increase the allocation to equities as he ages to maintain a constant overall exposure for risk—the theory does not say that there should be a large allocation to equity when he is old. As section 2 shows, an individual would generally prefer to hold more bonds, all else equal, if her labor income is correlated with equities. Thus, according to life-cycle theory, the stockbroker might have optimally held mostly bonds, but his (hopefully very small) equity portion would have gradually increased as he approached retirement.

While these examples illustrate the main intuition of labor income across the life cycle—that human capital is an asset and our holdings of financial assets should adjust to counter-balance the bond-ness or stock-ness of human capital—actual life-cycle models used by economists add several features. We have abstracted from leverage constraints, how a person consumes, and changing liabilities. Many students today graduate with large debt—so they begin with negative net wealth. What is most lacking, however, is that we treated the young person’s problem independently of the old person’s problem. In a true life-cycle model, the young person preemptively takes into account the reduced labor income when she is old. Our next (stylized) example rectifies this but shows that the same intuition holds.

Sovereign Wealth Fund Example

Sovereign wealth funds (SWFs) are tantamount to a retirement plan for a nation. Any SWF set up to diversify from (and outlast) commodity wealth is like a retirement fund. Just as the correlation of human capital returns with asset returns drives asset allocation over an investor’s life cycle, the correlation of commodity returns with asset returns should determine a SWF’s asset allocation.

Consider the following simple model of a SWF.²² There will be two periods: the first is when oil wealth is plentiful and the second when the resource has been mostly extracted. To simplify, we’ll abstract from a spending rule (consumption) and just maximize the expected utility of terminal wealth. (Recall this is not the same as maximizing the amount of money because the utility function trades off risk and return; see chapter 2.) There are two financial assets: risk-free bonds,

²²Some more formal models of SWF asset allocation are Scherer (2011) and Bodie and Brière (2013).